

Netflix Dashboard

Table of Contents

1. Introduction	1
1.1 Overview of Netflix and Its Content Library	1
1.2 Importance of Data Analysis and Visualization in Streaming	1
1.3 Objectives of the Project	1
2. Problem Statement	1
2.1 Challenges in Analyzing Netflix's Vast Content Library.....	1
2.2 Need for Global Distribution and Genre Analysis	2
2.3 Importance of Dynamic Data Visualizations	2
3. Literature Review.....	2
3.1 Studies on Streaming Service Data Analytics	2
3.2 SQL for Large-Scale Data Management	2
3.3 Tableau as a Tool for Business Intelligence.....	2
4. Methodology.....	3
4.1 Data Collection and Storage in PostgreSQL	3
4.2 PostgreSQL Database Setup.....	4
4.3 Data Cleaning and Preparation.....	6
4.4 Handling Missing Values	7
4.5 SQL Queries for Data Analysis	10
4.6 Tableau Dashboard Creation	11
4.7 Connecting Tableau to PostgreSQL	12
4.8 Building Visualizations	13
5. Results	14
5.1 Key Insights and Analysis of Netflix	14
5.2 Output.....	14
7. Future Work and Extensions	15
7.1 Real-Time Data Integration.....	15
7. Expanding User Behavior Analysis.....	15
7.3 Enhancing the Tableau Dashboard with Machine Learning Integration.....	15
8. Conclusion.....	15
8.1 Summary of Findings	15
8.2 Practical Applications of the Project's Results.....	15
9 . References	16

Introduction

Netflix has emerged as a global leader in the entertainment industry, offering an extensive catalog of movies, TV shows, and documentaries to millions of subscribers worldwide. With such a vast collection, it becomes increasingly important to analyze the structure and distribution of content to extract valuable insights. As the platform continues to expand, understanding how different types of content are distributed geographically and thematically can help Netflix optimize its catalog, while providing users with a personalized streaming experience.

This project aims to explore and visualize the Netflix catalog by utilizing PostgreSQL for data management and analysis, and Tableau for dynamic dashboard creation. The interactive dashboard serves as a tool for stakeholders to navigate through Netflix's diverse offerings, providing key insights such as the distribution of movies and TV shows, regional content availability, genre preferences, and content rating trends.

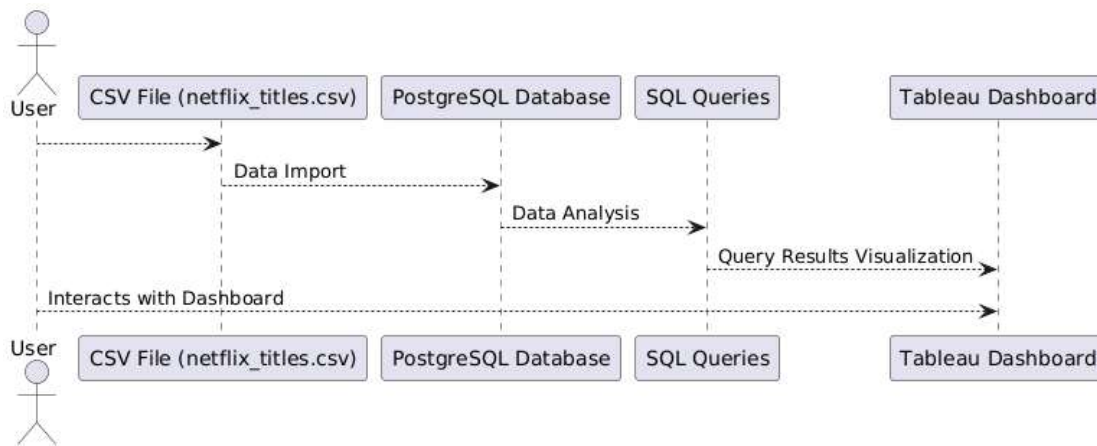
The significance of the project is to understand the composition of Netflix's catalog is crucial for content creators, marketers, and decision-makers in the entertainment industry. By providing data-driven insights, this analysis can help drive content strategies and marketing initiatives.

Problem Statement

The Netflix catalog presents an overwhelming amount of content, making it difficult for both users and analysts to identify patterns, preferences, and trends. There are several specific challenges:

- ❖ First, the massive volume of movies, TV shows, and other types of content leads to content overload, limiting the ability of users to make informed decisions about what to watch.
- ❖ Second, there is a lack of clear insights into the geographical distribution of content—users and stakeholders might not know how content varies across countries or regions.
- ❖ Third, Netflix caters to a diverse audience with varied preferences and age groups, making it important to classify content effectively based on genre, ratings, and release year. Furthermore, traditional static reports fail to offer the interactive, real-time insights that today's businesses and users need.

This project addresses these challenges by combining SQL analysis and interactive visualizations in Tableau, providing a platform to explore Netflix's catalog in a more structured, user-friendly manner.



Literature Review

The analysis of large datasets in the media and entertainment industry has become an increasingly popular area of research, as streaming platforms like Netflix, Hulu, and Amazon Prime collect vast amounts of data about their content and users. Several studies have explored the role of data analytics in understanding content distribution and user engagement. For example, previous research on media content analysis focuses on understanding trends across genres, release years, and geographical preferences. SQL, as a robust and versatile language for querying relational databases, has been used extensively in handling and analyzing large datasets like those from streaming platforms.

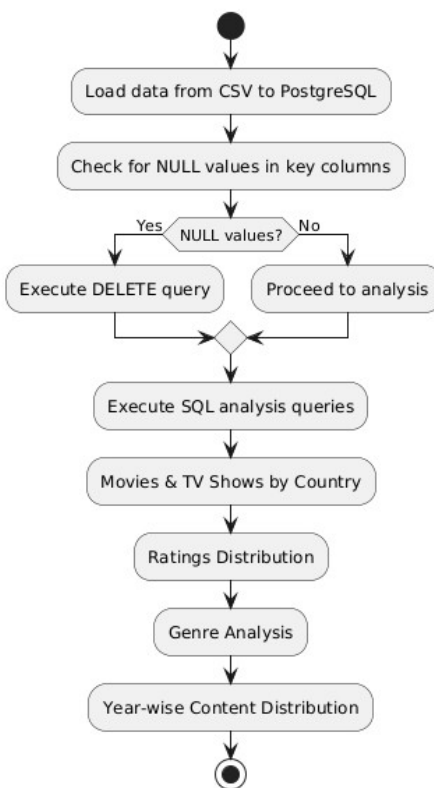
Similarly, the use of business intelligence tools, particularly Tableau, has gained prominence in the field of data visualization. Tableau's ability to provide interactive, visually engaging dashboards makes it a key tool for industries dealing with massive datasets. In this context, research also highlights the power of visual storytelling and its importance in delivering actionable insights to stakeholders who may not be familiar with complex data structures.

Furthermore, research on user preferences in streaming services often emphasizes the importance of content categorization—whether by region, genre, or rating—in predicting user behavior and enhancing user experience. Understanding these patterns helps platforms like Netflix optimize their content recommendation engines and informs their decision-making on what types of content to invest in for different markets.

Methodology

The methodology for this project involved several key steps to ensure thorough data analysis and visualization. The data was cleaned by removing any records with missing values in critical fields such as country and release_year. Following this, SQL queries were executed to analyze various aspects of the content library, including the distribution of titles by country, genre, ratings, and release year. These queries generated insights that were then imported into Tableau for visualization. Tableau was used to create an interactive dashboard that allows users to explore Netflix's content library dynamically, applying filters to view data by country, genre, ratings, and more.

As part of this project, the codebase and relevant documentation have been uploaded to a GitHub repository for version control and collaborative purposes. The repository can be accessed from [PoorviKesharwani/Tableau-Projects](https://github.com/PoorviKesharwani/Tableau-Projects).



Workflow

The Netflix dataset was first obtained from a publicly available source and contains key details about each title, including its genre, rating, country of production, and release year. To handle this data efficiently, we used PostgreSQL to store, clean, and query the dataset. The structured nature of PostgreSQL allowed for efficient execution of complex queries to filter, aggregate, and transform the data for further analysis.

Step 1: Install PostgreSQL and Create a Database

Before importing the dataset, PostgreSQL was installed on the local machine, and a database named netflix_db was created to store the Netflix data:

CREATE DATABASE Netflix Database;

Step 2: Table Creation

The following SQL script was executed to create the netflix_titles table with appropriate data types for each column:

```
CREATE TABLE netflix_titles (  
  show_id VARCHAR PRIMARY KEY,  
  type VARCHAR,  
  title VARCHAR,  
  director VARCHAR,  
  "cast" TEXT,  
  country VARCHAR,  
  date_added DATE,  
  release_year INT,  
  rating VARCHAR,  
  duration VARCHAR,  
  listed_in TEXT,  
  description TEXT  
);
```

CREATE TABLE

Query returned successfully in 50 msec.

	show_id [PK] character varying	type character varying	title character varying	director character varying	cast_list text	country character varying	date_added date	release_year integer	rating character varying	duration character varying	listed_in text	description text
1	81145628	Movie	Norm of the North: King ...	Richard Finn, Tim Ma...	Alan Marriott, A...	United States, India, ...	2019-09-09	2019	TV-PG	90 min	Children & Family M...	Before pl...
2	80125979	Movie	#realityhigh	Fernando Lebrija	Nesta Cooper, ...	United States	2017-09-08	2017	TV-14	99 min	Comedies	When ner...
3	70304989	Movie	Automata	Gabe Ibáñez	Antonio Bander...	Bulgaria, United Stat...	2017-09-08	2014	R	110 min	International Movies,...	In a dysto...
4	80164077	Movie	Fabrizio Copano: Solo pie...	Rodrigo Toro, Francis...	Fabrizio Copano	Chile	2017-09-08	2017	TV-MA	60 min	Stand-Up Comedy	Fabrizio C...
5	70304990	Movie	Good People	Henrik Ruben Genz	James Franco, ...	United States, United...	2017-09-08	2014	R	90 min	Action & Adventure, ...	A struggli...
6	70299204	Movie	Kidnapping Mr. Heineken	Daniel Alfredson	Jim Sturgess, S...	Netherlands, Belgiu...	2017-09-08	2015	R	95 min	Action & Adventure, ...	When bee...
7	80057969	Movie	Love	Gaspar Noé	Karl Glusman, K...	France, Belgium	2017-09-08	2015	NR	135 min	Cult Movies, Dramas,...	A man in ...
8	80060297	Movie	Manhattan Romance	Tom O'Brien	Tom O'Brien, Ka...	United States	2017-09-08	2014	TV-14	98 min	Comedies, Independ...	A filmma...
9	80046728	Movie	Moonwalkers	Antoine Bardou-Jacq...	Ron Perlman, R...	France, Belgium	2017-09-08	2015	R	96 min	Action & Adventure, ...	A brain-a...
10	70304988	Movie	Stonehearth Asylum	Brad Anderson	Kate Beckinsal...	United States	2017-09-08	2014	PG-13	113 min	Horror Movies, Thrill...	In 1899, a...
11	80057700	Movie	The Runner	Austin Stark	Nicolas Cage, S...	United States	2017-09-08	2015	R	90 min	Dramas, Independen...	A New Orl...
12	80045922	Movie	6 Years	Hannah Fidell	Taissa Farniga...	United States	2015-09-08	2015	NR	80 min	Dramas, Independen...	As a volat...
13	70241607	Movie	Laddaland	Sopon Sukdapisit	Saharat Sangka...	Thailand	2018-09-07	2011	TV-MA	112 min	Horror Movies, Intern...	When a fa...
14	80988892	Movie	Next Gen	Kevin R. Adams, Joe ...	John Krasinski, ...	China, Canada, Unite...	2018-09-07	2018	TV-PG	106 min	Children & Family M...	When lon...
15	80239639	Movie	Sierra Burgess Is A Loser	Ian Samuels	Shannon Purser...	United States	2018-09-07	2018	PG-13	106 min	Comedies, Romantic...	A wrong...
16	80159586	Movie	The Most Assassinated ...	Franck Ribière	Anna Mouglalis...	Belgium, United King...	2018-09-07	2018	TV-MA	102 min	Dramas, Internationa...	In 1930s ...
17	80152447	Movie	Cézanne et moi	Daniele Thompson	Guillaume Cane...	Belgium, France	2017-09-07	2016	R	114 min	Dramas, Independen...	This histo...
18	81154455	Movie	Article 15	Anubhav Sinha	Ayushmann Kh...	India	2019-09-06	2019	TV-MA	125 min	Dramas, Internationa...	The grim ...
19	81052275	Movie	Ee Nagarani Emaiindi	Tharun Bhascker	Vishwaksen Na...	India	2019-09-06	2018	TV-14	133 min	Comedies, Internatio...	In Goa an...
20	80058026	Movie	Hell and Back	Tom Gianas, Ross R. ...	Nick Swardson,...	United States	2018-09-06	2015	R	86 min	Action & Adventure, ...	When bes...
21	70303496	Movie	PK	Rajkumar Hirani	Aamir Khan, An...	India	2018-09-06	2014	TV-14	146 min	Comedies, Dramas, I...	Aamir Kh...
22	80162141	Movie	Hard Tide	Robert Osman, Natha...	Nathanael Wise...	United Kingdom	2017-09-06	2015	TV-MA	79 min	Dramas, Independen...	A drug de...

The table was structured to hold metadata about each Netflix title, including its type (Movie or TV Show), director, cast, country of origin, date added to the platform, release year, rating (e.g., TV-MA, PG-13), and genre classification. The primary key for the table was the show_id.

Step 3: Data Import into PostgreSQL

To add the table, the CSV file containing Netflix's catalog was imported using the COPY command, which efficiently loads large datasets into PostgreSQL:

```
COPY netflix_titles FROM '/path/to/netflix_titles.csv' DELIMITER ',' CSV HEADER;
```

COPY 6234

Query returned successfully in 101 msec.

Data Cleaning and Preparation

Once the data was loaded into PostgreSQL, several steps were followed for data cleaning. Missing values in critical columns such as country and release_year were identified and removed, as these could skew the analysis. This cleaning process was automated using SQL commands such as DELETE to remove rows with NULL values.

Step 4: Handling Missing Values

To remove rows where the country or release_year columns had NULL values, the following query was executed:

```
DELETE FROM netflix_titles  
WHERE country IS NULL OR release_year IS NULL;
```

	show_id [PK] character varying	type character varying	title character varying	director character var	cast_list text	country character var	date_added date	release_year integer	rating character var	duration character var	listed_in text	description text
1	80169755	Movie	Joaquín Reyes: U...	José Mig...	Joaquín Reyes	[null]	2017-09-08	2017	TV-MA	78 min	Stand-Up Comedy	Comedian and celebri...
2	80182480	Movie	Krish Trish and B...	[null]	Damandeep Singh...	[null]	2017-09-08	2009	TV-Y7	58 min	Children & Family...	A team of minstrels, l...
3	80182483	Movie	Krish Trish and B...	Munjal Sh...	Damandeep Singh...	[null]	2017-09-08	2013	TV-Y7	62 min	Children & Family...	An artisan is cheated ...
4	80182596	Movie	Krish Trish and B...	Munjal Sh...	Damandeep Singh...	[null]	2017-09-08	2016	TV-Y	65 min	Children & Family...	A cat, monkey and do...
5	80182482	Movie	Krish Trish and B...	Tilak Shet...	Damandeep Singh...	[null]	2017-09-08	2012	TV-Y7	61 min	Children & Family...	In three comic-strip-st...
6	80182597	Movie	Krish Trish and B...	Tilak Shet...	Rishi Gambhir, Sm...	[null]	2017-09-08	2017	TV-Y7	65 min	Children & Family...	A cat, monkey and do...
7	80182481	Movie	Krish Trish and B...	[null]	Damandeep Singh...	[null]	2017-09-08	2010	TV-Y7	58 min	Children & Family...	Animal minstrels narr...
8	80182621	Movie	Krish Trish and B...	Munjal Sh...	Damandeep Singh...	[null]	2017-09-08	2013	TV-Y7	60 min	Children & Family...	The consequences of ...
9	80244601	TV Show	Castle of Stars	[null]	Chaiyapol Pupart, ...	[null]	2018-09-07	2015	TV-14	1 Season	International TV ...	As four couples with d...
10	80190843	TV Show	First and Last	[null]	[null]	[null]	2018-09-07	2018	TV-MA	1 Season	Docuseries	Take an intimate look ...
11	80221550	TV Show	Archibald's Next ...	[null]	Tony Hale, Rosam...	[null]	2019-09-06	2019	TV-Y7	1 Season	Kids' TV, TV Com...	Happy-go-lucky chicke...
12	81113928	Movie	Care of Kanchara...	Maha Ven...	Subba Rao Vepad...	[null]	2019-09-06	2018	TV-14	142 min	Comedies, Dram...	From a schoolboy's cr...
13	81132437	Movie	Kill Me if You Dare	Şenol Sön...	Murat Boz, Seda B...	[null]	2019-09-06	2019	TV-14	100 min	Comedies, Intern...	Bogged down with mo...
14	80163293	TV Show	Mak Cun	[null]	Erma Fatima, Rahi...	[null]	2017-09-29	2015	TV-PG	1 Season	International TV ...	A woman with an entr...
15	81176188	Movie	American Factory...	[null]	President Barack ...	[null]	2019-09-05	2019	TV-G	10 min	Documentaries	Barack and Michelle O...
16	80184358	TV Show	Lovesick	[null]	Kongyingyong Ch...	[null]	2018-09-03	2014	TV-14	1 Season	International TV ...	Love and academics g...
17	80119349	Movie	Out of Thin Air	Dylan Ho...	[null]	[null]	2017-09-29	2017	TV-14	85 min	Documentaries, I...	Picking up 40 years af...
18	80180449	TV Show	Satu Hari	[null]	Sara Ali, Remy Ish...	[null]	2017-09-29	2014	TV-14	1 Season	International TV ...	A man haunted by his ...
19	81053893	Movie	Cultivating the Se...	[null]	Mana Yas...	[null]	2019-09-28	2019	TV-G	45 min	Documentaries, I...	An in-depth look at th...
20	81053892	Movie	TUNA GIRL	[null]	Mana Yas...	Fuka Koshiba, To...	2019-09-28	2019	TV-PG	90 min	Dramas, Internati...	A clumsy but cheerful ...
21	80218107	TV Show	Dragons: Rescue ...	[null]	Nicolas Cantu, Br...	[null]	2019-09-27	2019	TV-Y	1 Season	Kids' TV, TV Com...	Twins Dak and Leyla a...
22	80220715	TV Show	Skylines	[null]	Edin Hasanovic, P...	[null]	2019-09-27	2019	TV-MA	1 Season	Crime TV Shows, ...	A hip-hop producer ge...

DELETE 0

Query returned successfully in 34 msec.

After cleaning the dataset, we conducted a thorough analysis using SQL queries to derive insights about content distribution. Key queries included aggregations to understand the distribution of movies versus TV shows, the prevalence of different content ratings (such as TV-MA, PG-13), and the geographical spread of titles across different countries. Additionally, SQL was used to group and count titles by release year, offering insights into Netflix's content growth over time.

	show_id [PK] character varying	type character varying	title character varying	director character var	cast_list text	country character var	date_added date	release_year integer	rating character var	duration character var	listed_in text	description text
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13	81132437	Movie	Kill Me if You Dare	Şenol Sön...	Murat Boz, Seda B...	[null]	2019-09-06	2019	TV-14	100 min	Comedies, Intern...	Bogged down with mo...
14	80163293	TV Show	Mak Cun	[null]	Erma Fatima, Rahl...	[null]	2017-09-29	2015	TV-PG	1 Season	International TV ...	A woman with an entr...
15	81176188	Movie	American Factory...	[null]	President Barack ...	[null]	2019-09-05	2019	TV-G	10 min	Documentaries	Barack and Michelle O...
16	80184358	TV Show	Lovesick	[null]	Kongyingyong Ch...	[null]	2018-09-03	2014	TV-14	1 Season	International TV ...	Love and academics g...
17	80119349	Movie	Out of Thin Air	Dylan Ho...	[null]	[null]	2017-09-29	2017	TV-14	85 min	Documentaries, I...	Picking up 40 years af...
18	80180449	TV Show	Satu Hari	[null]	Sara Ali, Remy Ish...	[null]	2017-09-29	2014	TV-14	1 Season	International TV ...	A man haunted by his ...
19	81053893	Movie	Cultivating the Se...	Mana Yas...	[null]	[null]	2019-09-28	2019	TV-G	45 min	Documentaries, I...	An in-depth look at th...
20	81053892	Movie	TUNA GIRL	Mana Yas...	Fuka Koshiba, To...	[null]	2019-09-28	2019	TV-PG	90 min	Dramas, Internati...	A clumsy but cheerful ...
21	80218107	TV Show	Dragons: Rescue ...	[null]	Nicolas Cantu, Br...	[null]	2019-09-27	2019	TV-Y	1 Season	Kids' TV, TV Com...	Twins Dak and Leyla a...
22	80220715	TV Show	Skylines	[null]	Edin Hasanovic, P...	[null]	2019-09-27	2019	TV-MA	1 Season	Crime TV Shows, ...	A hip-hop producer ge...

Data Analysis in PostgreSQL

With the cleaned dataset, we proceeded to analyze key aspects of Netflix's content library. The following SQL queries were executed to extract insights:

Step 5: Analyzing Global Distribution of Titles

This analyses involved understanding how Netflix's content is distributed across different countries. The query below counts the total number of titles available in each country:

```
SELECT country, COUNT(*) as total_titles
FROM netflix_titles
WHERE country IS NOT NULL
GROUP BY country
ORDER BY total_titles DESC;
```

	country character varying 🔒	total_titles bigint 🔒
1	United States	1326
2	India	707
3	United Kingdom	152
4	Canada	78
5	Spain	72
6	Turkey	60
7	Japan	54
8	Mexico	51
9	Hong Kong	48
10	France	46

This query provides a breakdown of titles by country, helping to identify Netflix's strongest markets in terms of content availability. The results from this query were later visualized on a map in Tableau, offering an intuitive view of Netflix's global content distribution.

Step 6: Content Classification by Rating

This query examined how Netflix content is classified by ratings, giving insights into the types of content offered on the platform. This was particularly useful for understanding whether Netflix's catalog is skewed toward more mature content (TV-MA) or family-friendly programming (TV-G):

```
SELECT rating, COUNT(*) as total_titles  
FROM netflix_titles  
WHERE rating IS NOT NULL  
GROUP BY rating  
ORDER BY total_titles DESC;
```

	rating character varying 🔒	total_titles bigint 🔒
1	TV-MA	1189
2	TV-14	917
3	R	501
4	TV-PG	358
5	PG-13	278
6	PG	176
7	NR	175
8	TV-G	54
9	TV-Y7	48
10	G	35
11	TV-Y	24
12	TV-Y7-FV	11
13	UR	7
14	NC-17	1

The results from this query were used to create a bar chart in Tableau, displaying the distribution of content ratings.

Step 7: Movies vs TV Shows Distribution

To better understand the structure of Netflix's library, the following query was executed to compare the number of movies and TV shows:

```
SELECT type, COUNT(*) as total_count
FROM netflix_titles
GROUP BY type;
```

	type character varying 🔒	total_count bigint 🔒
1	Movie	3682
2	TV Show	96

This query outputs the total number of movies and TV shows, which was later visualized as a pie chart in Tableau, helping users understand the ratio between these two content types.

Step 8: Top 10 Genres by Number of Titles

Genres are a critical component in determining user preferences. To identify the most popular genres on Netflix, the following query was used to rank the top 10 genres based on the number of titles available:

```
SELECT listed_in, COUNT(*) as total_titles
FROM netflix_titles
GROUP BY listed_in
ORDER BY total_titles DESC
LIMIT 10;
```

The results were visualized in Tableau as a horizontal bar chart, showcasing the top 10 genres in Netflix's library.

	listed_in text 🔒	total_titles bigint 🔒
1	Dramas, International Movies	237
2	Stand-Up Comedy	234
3	Dramas, Independent Movies, International Mov...	184
4	Comedies, Dramas, International Movies	168
5	Documentaries	136
6	Children & Family Movies, Comedies	118
7	Comedies, International Movies	110
8	Dramas, International Movies, Romantic Movies	103
9	Action & Adventure, Dramas, International Movies	99
10	Comedies, International Movies, Romantic Movi...	93

Step 9: Titles Released by Year

Finally, the growth of Netflix's content library over time was analyzed using the following query, which groups titles by their release year:

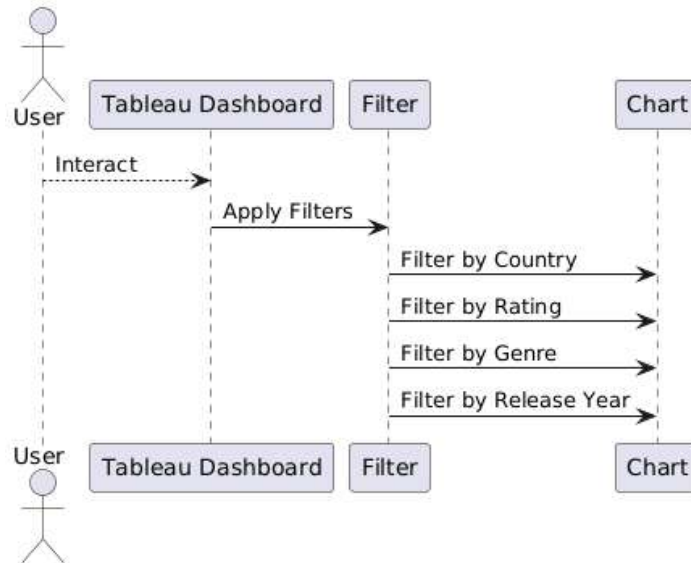
```
SELECT release_year, type, COUNT(*) as total_titles
FROM netflix_titles
GROUP BY release_year, type
ORDER BY release_year;
```

	release_year integer	total_titles bigint
1	1942	1
2	1944	1
3	1945	1
4	1946	2
5	1947	1
6	1954	1
7	1955	1
8	1956	1
9	1958	2
10	1959	1
11	1960	3
12	1962	3
13	1964	1
14	1965	2
15	1966	1
16	1967	2
17	1968	3
18	1969	2
19	1970	1
20	1971	3
21	1972	3
22	1973	9
23	1974	7
24	1975	5

This query allowed us to visualize the trend in the number of movies and TV shows released each year, illustrating Netflix's evolving content strategy. The results were displayed in Tableau as an area chart.

Tableau Dashboard Creation

Once the data was analyzed, it was imported into Tableau for visualization. A live connection between PostgreSQL and Tableau was established to ensure that any updates to the dataset would be reflected immediately in the dashboard. Tableau was then used to create various charts and maps to visualize the key findings, such as the number of titles by country, distribution of genres, and the evolution of content over the years.



Step 10: Connecting Tableau to PostgreSQL

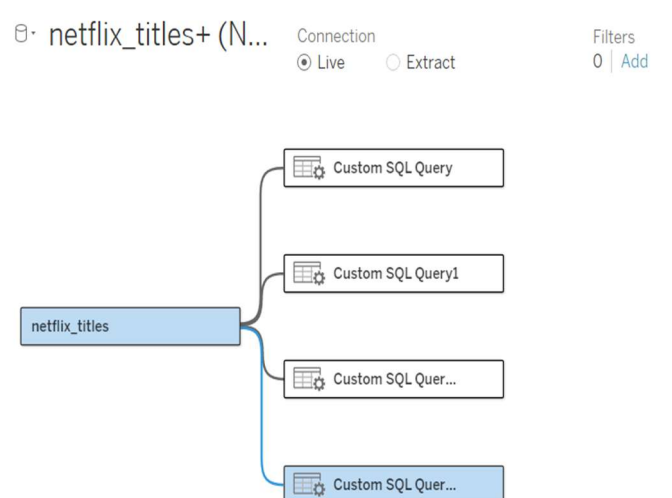
To visualize the data in Tableau, a live connection was established with the PostgreSQL database. This allows Tableau to fetch the most up-to-date data whenever new content is added to Netflix's catalog.

- Open Tableau and go to *Data > Connect to Data*.
- Select PostgreSQL as the data source.
- Enter the PostgreSQL server credentials, including the server address, port, database name, username, and password.

The screenshot shows the PostgreSQL connection configuration window in Tableau. The window has two tabs: 'General' and 'Initial SQL'. The 'General' tab is active, showing the following fields:

- Server: 127.0.0.1
- Port: 5432
- Database: Netflix Database
- Authentication: Username and Password
- Username: postgres
- Password: Optional
- ☐ Require SSL

A 'Sign In' button is located at the bottom right of the window.

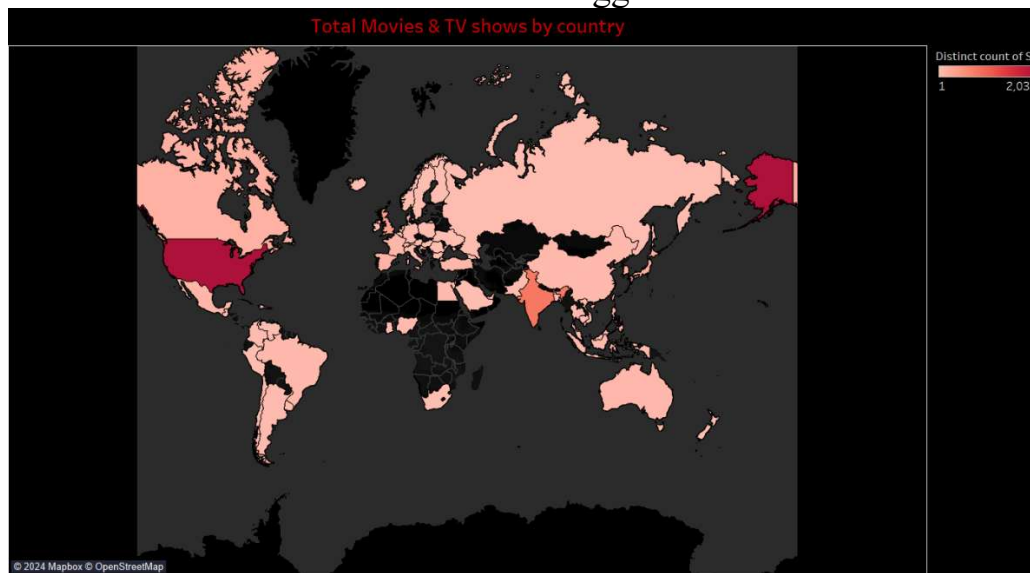


Step 11: Creating Visualizations

Tableau was used to visualize the results. Tableau's rich set of features, including interactive filters and dynamic visualizations, made it the ideal tool for presenting the data in an intuitive format.

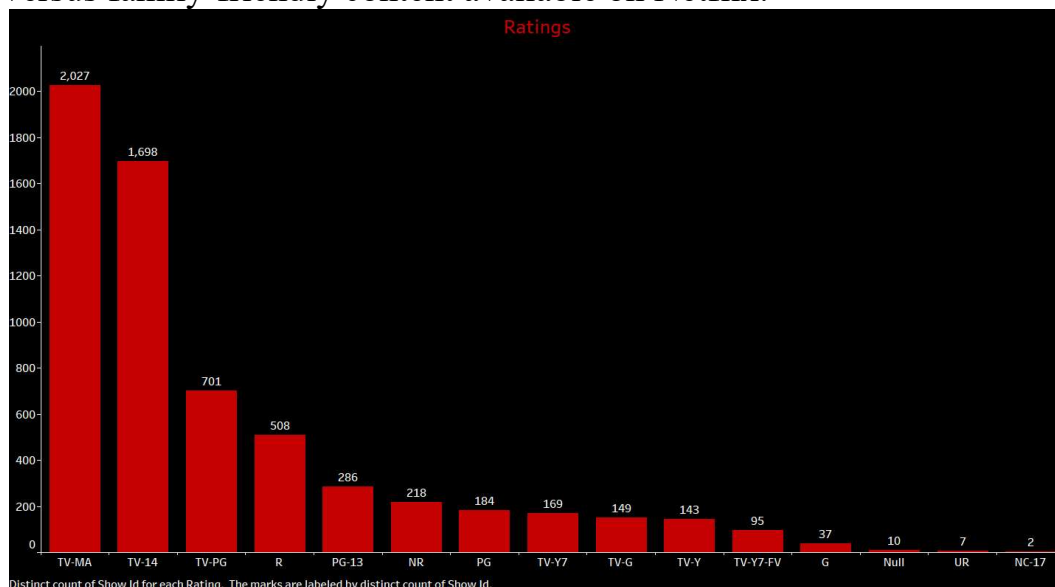
1. Map Visualization (Movies & TV Shows by Country)

To visualize the distribution of Netflix content across the globe, a **map** was created in Tableau. The results from the country query were plotted using geographic data points, where each country was represented by a circle scaled to reflect the number of titles available. Filters were added to allow users to toggle between movies and TV shows.



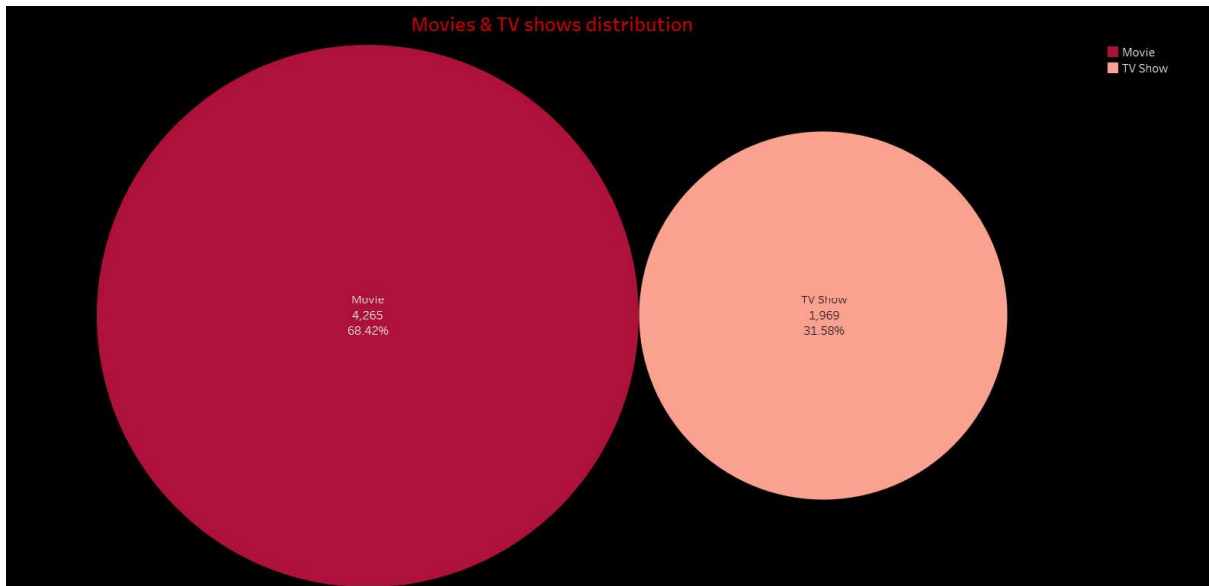
2. Ratings Distribution Bar Chart

Using the results of the rating query, a **bar chart** was created to show the distribution of titles across different ratings. This chart helps users understand the proportion of mature content versus family-friendly content available on Netflix.



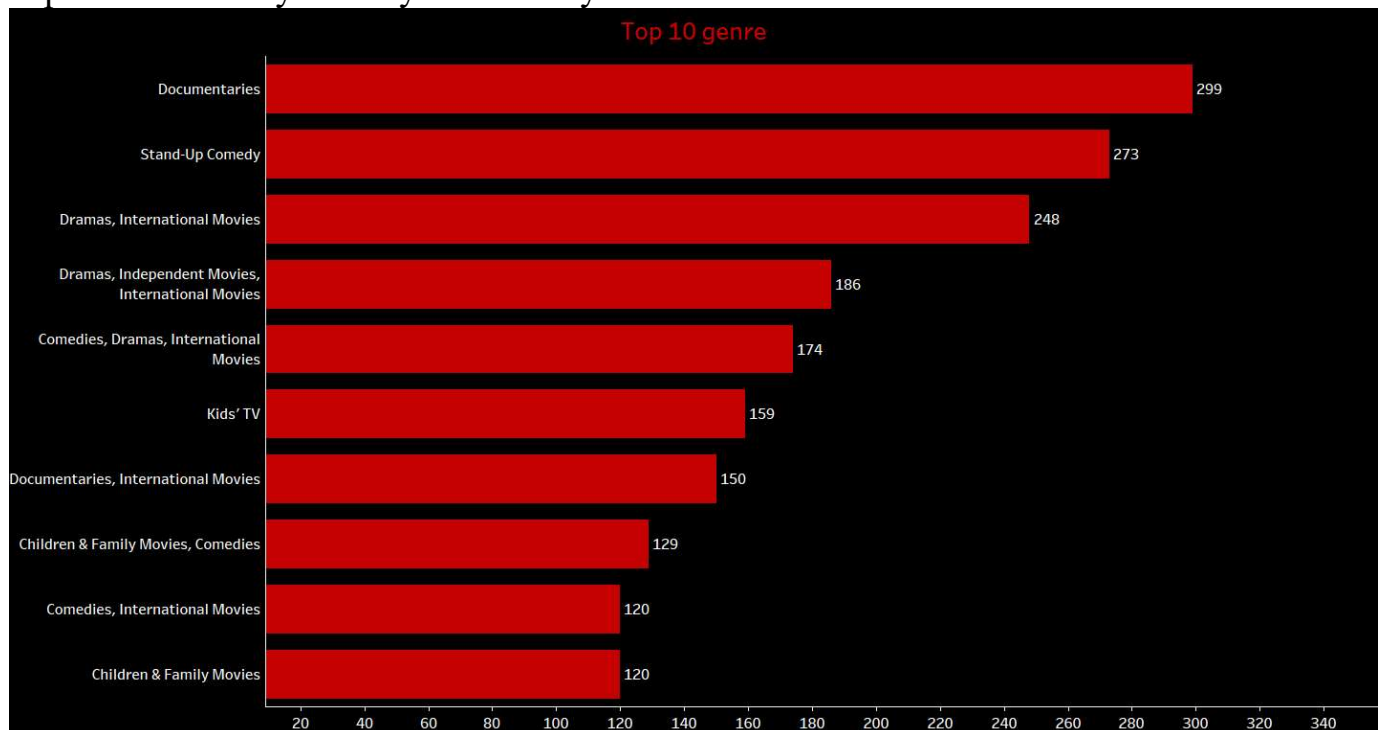
3. Movies vs TV Shows Pie Chart

A pie chart was designed to visually represent the ratio of movies to TV shows on Netflix. This chart was interactive, allowing users to filter other visualizations based on their selection of movies or TV shows.



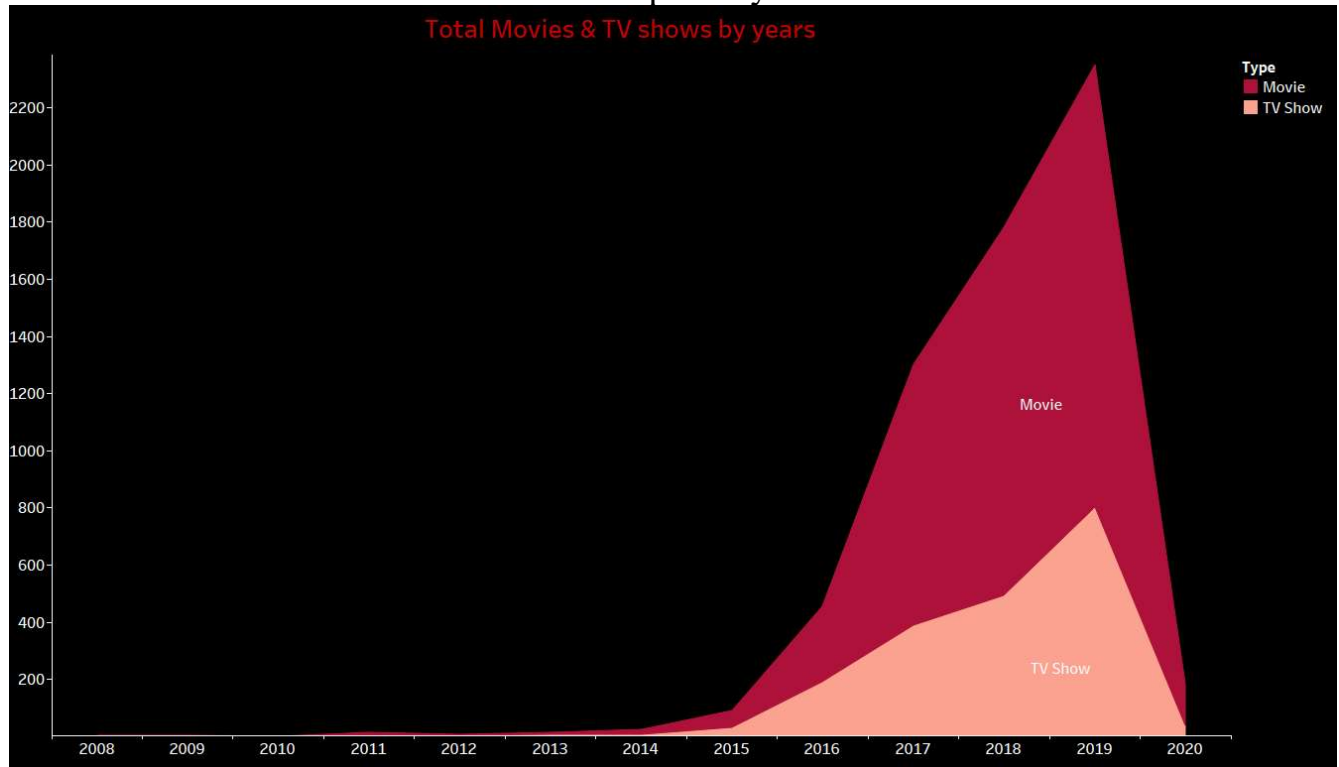
4. Genre Analysis Horizontal Bar Chart

The top 10 genres on Netflix were displayed using a **horizontal bar chart**, allowing users to explore which genres dominate the platform's library. Filters were added to explore the data by country or release year.



5. Area Chart (Total Titles Released by Year)

To visualize the growth of Netflix's content library, an **area chart** was created, showing the total number of titles released each year. Users could filter the chart to explore the release trends for movies and TV shows separately.



Step 12: Adding Interactivity and Filters

To enhance the user experience, several interactive features were added to the Tableau dashboard:

- **Filters** for type, release_year, country, and rating, allowing users to dynamically explore different aspects of the Netflix catalog.
- **Tooltips** providing additional information about specific data points, such as the exact number of titles for a given genre or country.

The final dashboard was a comprehensive, interactive tool that allowed users to explore Netflix's content library from multiple perspectives.

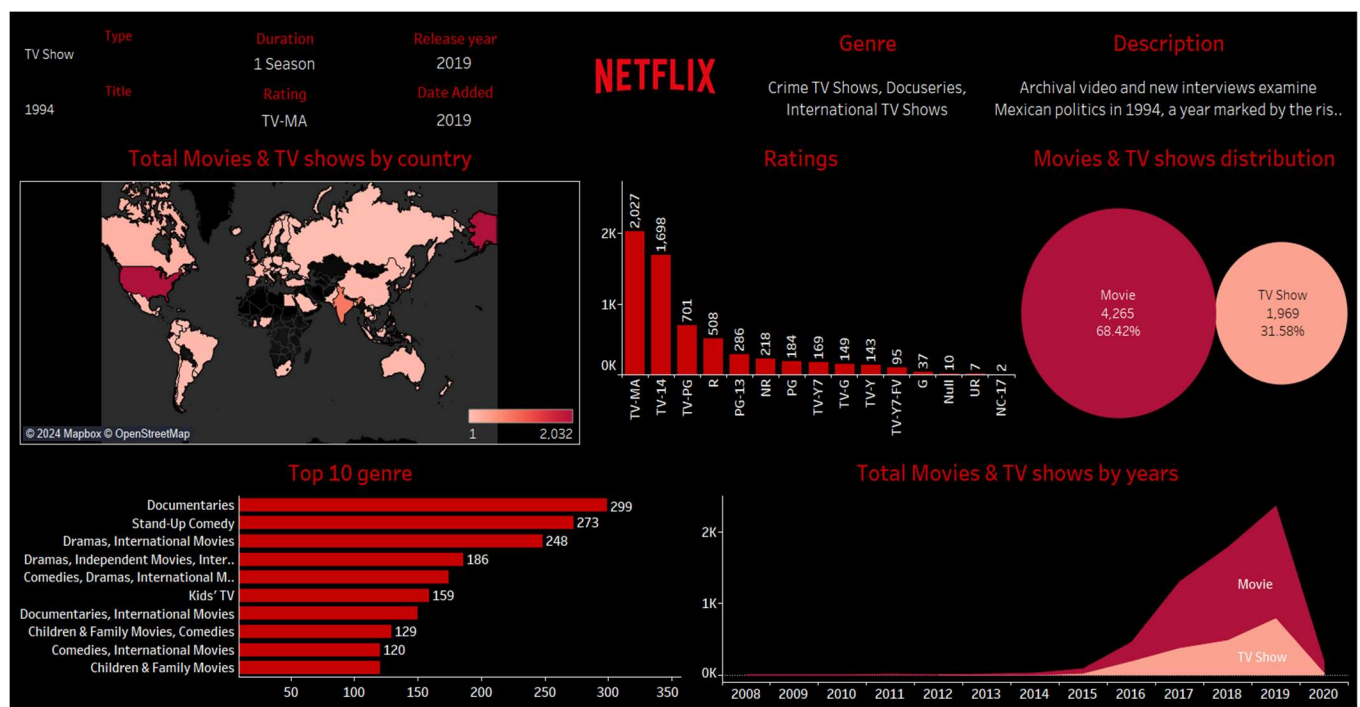
Type	Duration	Release year
TV Show	1 Season	2019
Title	Rating	Date Added
1994	TV-MA	2019

Genre	Description
Crime TV Shows, Docuseries, International TV Shows	Archival video and new interviews examine Mexican politics in 1994, a year marked by the ris..

Results

The analysis and visualization revealed several key insights about Netflix's content distribution. First, the geographical distribution showed that the United States and India have the most titles, reflecting Netflix's strong presence in these markets. A key feature of this dashboard is the ability for users to apply various filters, allowing them to customize their view of the data based on specific attributes such as content type and title.

The Tableau dashboard includes interactive filters for **Type** (Movies vs. TV Shows) and specific **Titles**, allowing users to explore Netflix's content more dynamically. The Type filter enables users to toggle between movies and TV shows, providing insights into the distribution and characteristics of each format, such as how they are spread across countries, genres, and ratings. The Title filter allows users to focus on individual shows or movies, providing detailed information such as release year, genre, and country of origin for each title. These filters enhance the dashboard's usability by enabling more personalized, in-depth data exploration and comparison between different content types and specific titles.



Additionally, genre analysis showed that Documentaries, Stand-Up Comedy, and International Movies were among the most popular genres on the platform. This indicates that Netflix's content strategy is expanding beyond mainstream genres to attract more niche audiences. The time-series analysis revealed that Netflix has significantly increased its content production since 2015, particularly in terms of original content.

Future Scope

Moving forward, the project can be expanded in several ways. One area of focus could be integrating real-time data feeds, ensuring that the dashboard remains up-to-date as Netflix continues to add new content. Another promising direction is to incorporate user behavior data, such as viewing habits and ratings, which could provide insights into how audiences engage with different types of content.

- **Enhanced Data Integration and Analytics:** Future developments could focus on integrating additional data sources and implementing machine learning algorithms to enable predictive analytics, providing deeper insights and more comprehensive analyses.
- **User Experience and Collaboration Improvements:** Enhancing the user interface for better intuitiveness and optimizing the project for mobile devices would improve accessibility. Additionally, incorporating real-time collaboration features would foster teamwork and streamline communication among users.
- **Automated Reporting and Performance Optimization:** Implementing automated reporting functionalities would facilitate regular updates for stakeholders. Concurrently, efforts to optimize data processing speeds will ensure efficient handling of larger datasets.

Furthermore, predictive analytics techniques could be applied to forecast content trends, helping Netflix anticipate future user preferences and make data-driven decisions about content acquisition and production. Machine learning algorithms could also be integrated to enhance the recommendation system, tailoring content suggestions based on a more granular understanding of user behavior.

Conclusion

This project demonstrated the effectiveness of combining PostgreSQL's data management and querying capabilities with Tableau's visualization tools to analyze Netflix's content library. The results offer valuable insights into Netflix's global content distribution, content ratings, and genre preferences, providing a data-driven understanding of the platform's content strategy.

The interactive Tableau dashboard provides a powerful tool for stakeholders to explore Netflix's content library in real-time, making it easier to identify trends and patterns that can inform decision-making. The project also highlights the importance of structured data analysis in optimizing content recommendations and improving user experiences on streaming platforms.

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